Seminar in Laboratory of Visual Intelligence and Pattern Analysis (VIPA)

An Introduction to Deep Image Aesthetics

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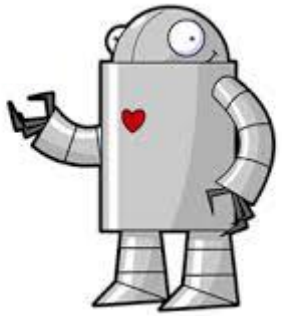
Outline

- Problem Statement
- Development
- Methods
  - Traditional Methods
  - **Deep Methods**
- Conclusions and Future Work

Problem Statement

- (**Photographic**) Image Aesthetics Assessment
  - Computationally **distinguishing** high-quality photos form low-quality ones based on photographic rules, typically in the form of:
  - **Binary Classification.** → **Classification Problem**
  - **Quality Scoring**. → **Regression Problem**



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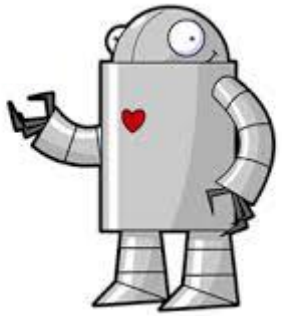
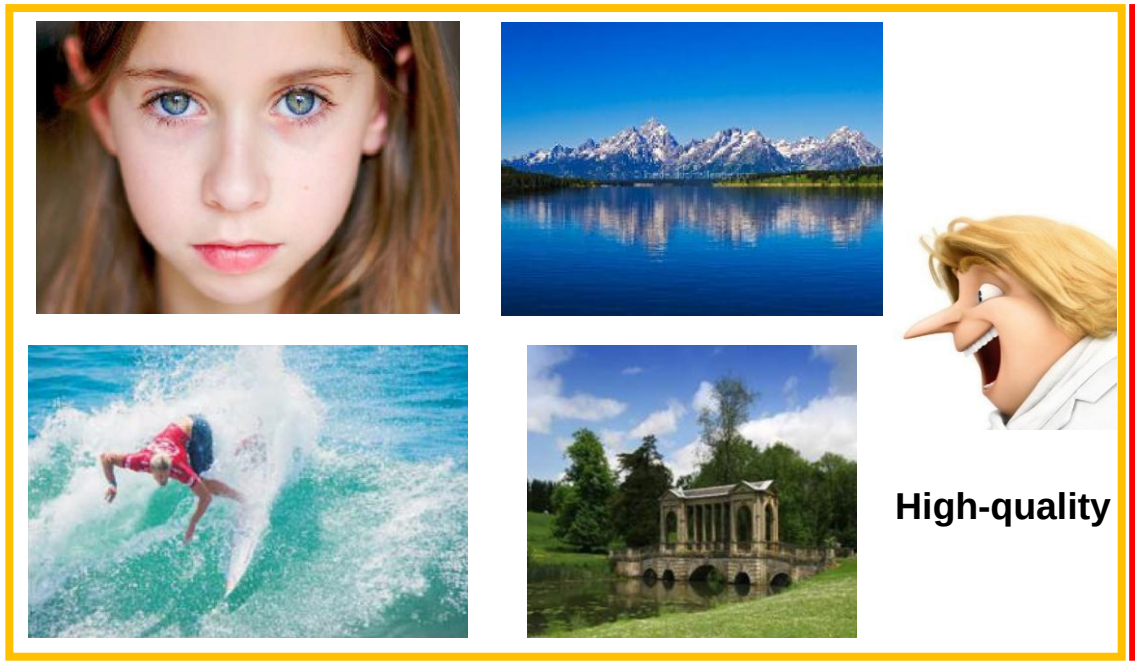
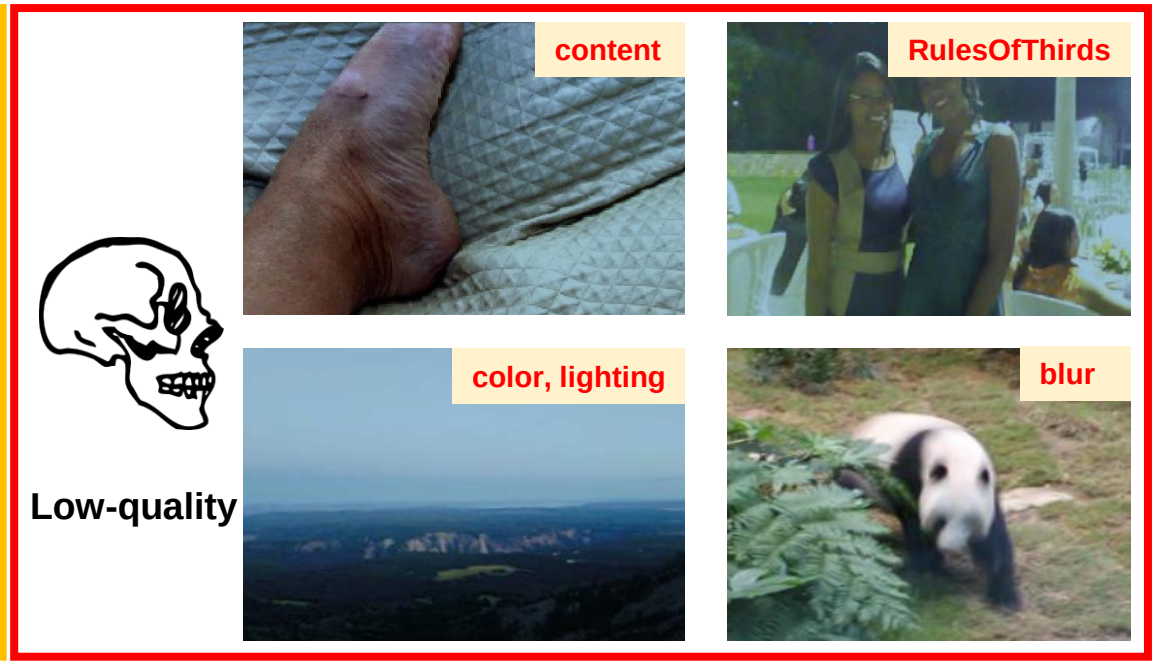
- Deep Image Aesthetics
  - Deep learning based image aesthetics assessment.



Problem Statement

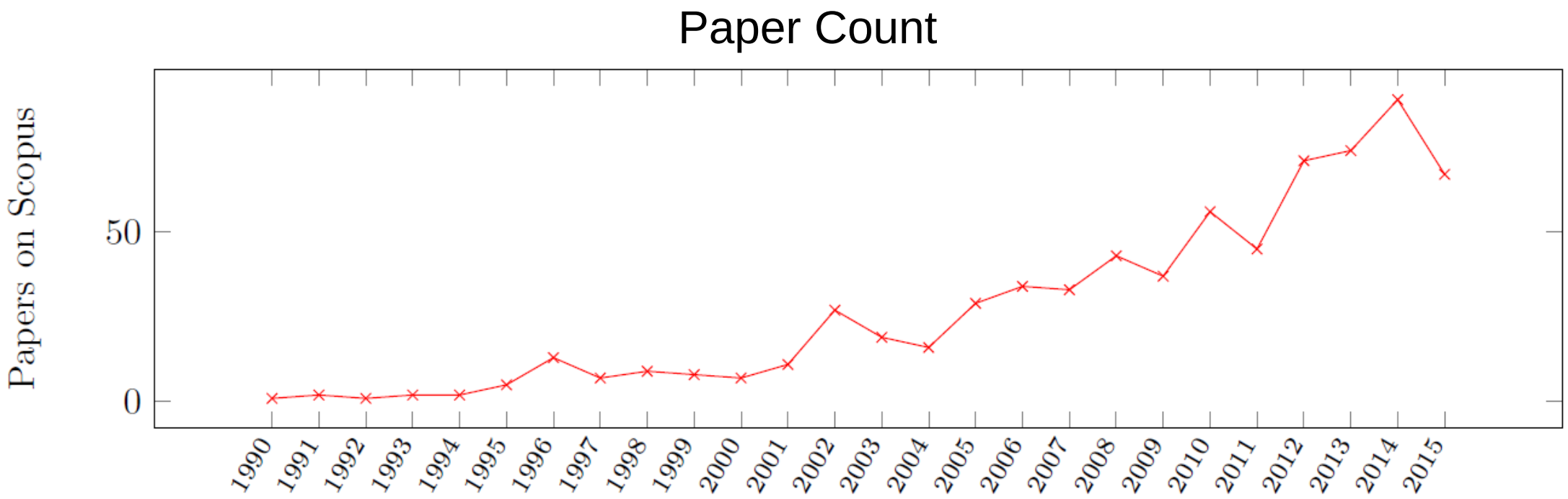
- Examples of High-Quality (Photographic) Images and Low-Quality Images.





Development

- **[1]:** “If we dig more on Scopus data, we find that the majority of publications comes from:
  - Asia (National University of Singapore, University Tenaga Nasional and Zhejiang University)
  - and North America (Simon Fraser University, Carnegie Mellon University, and Georgia Institute of Technology).”



[1] Spathis, D. (2016). Photo-Quality Evaluation based on Computational Aesthetics: Review of Feature Extraction Techniques. *arXiv preprint arXiv:1612.06259*.

Methods

- Framework

Input Image

Feature Extraction

Handcrafted Features

- Simple Image Features
- Image Composition Features
- General-Purpose Features
- Task-Specific Features

Deep Features

- Generic Deep Features
- Learned Aesthetics Deep Features

Traditional Methods

Deep Methods

Decision

Classification

Regression

[2] Deng, Y., Loy, C. C., & Tang, X. (2017). Image aesthetic assessment: An experimental survey. IEEE Signal Processing Magazine, 34(4), 80-106.

Deep Aesthetic Methods

- **2017**
  - **ICCV:** **Personalized** Image Aesthetics
  - **ICCV:** Deep **Cropping** via Attention Box Prediction and Aesthetics Assessment
  - **CVPR:** A-Lamp: Adaptive **Layout-Aware** Multi-Patch Deep Convolutional Neural Network for Photo Aesthetic Assessment
  - **TIP:** Deep Aesthetic Quality Assessment with **Semantic** Information
- **2016**
  - **CVPR:** **Composition-preserving** Deep Photo Aesthetics Assessment
  - **ECCV:** Photo Aesthetics Ranking Network with **Attributes and Content** Adaptation
  - **ACM MM:** Joint Image and **Text Representation** for Aesthetics Analysis
- **2015**
  - **ICCV:** Deep **Multi-Patch** Aggregation Network for Image Style, Aesthetics, and Quality Estimation

Deep Aesthetic Methods

- **Approach:** In general, exploit **Multi-task CNN**
  - **Multi-task** involves *aesthetic assessment*, *semantic content prediction*, *attribute prediction, etc.*

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Sampling Strategies:
1. Sampling pairs of images with a relatively large difference in their average aesthetic scores.
2. Sample image pairs that have been scored by the same individual.

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  - **Dataset:** AVA, AADB, etc. (see more in the next slide)

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Public Dataset

- Data label example:

Score **Attribute**

BalancElement
ColorHarmony
InterestContent
ShallowDOF
Good Lighting
ObjectEmphsis
RuleOfThirds
VividColor

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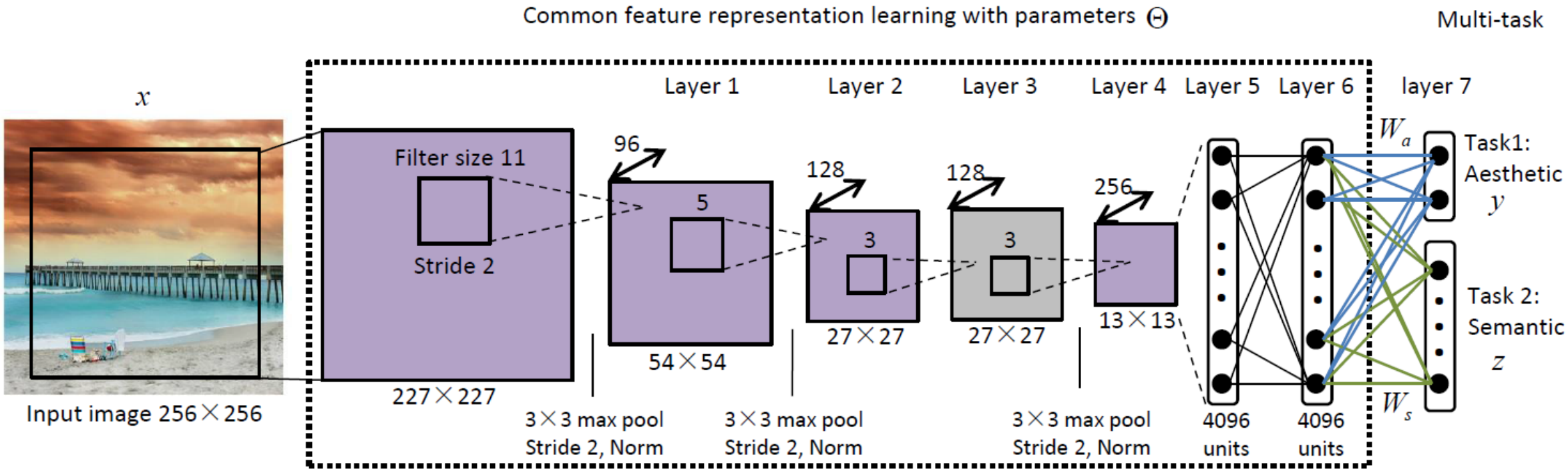
Public Dataset

- A summary of current datasets:

NAME	TOTAL IMG #	RATING PEOPLE # PER IMG	DESCRIPTION
Photo.Net	20,278	> 10	1) The score is from 0 to 7.
CUHK-PQ	17,690	8 - 10	1) **Binary** label. 2) Has **semantic** tags.
AVA	~25,000	78 - 549	1) The score is from 1 to 10. 2) Has **semantic** tags and **attribute** tags.
AADB	10,1000	5	1) **Five** workers annotate all the images. 2) Has **semantic** tags and **attribute** tags. 3) Attribute tags are **confidence scores**.
FLICKR-AES	40,000	5	1) The score is from 1 to 5.

Selected Deep Aesthetic Methods

- Multi-task Convolutional Neural Network (MTCNN)
  - **Overview**



[3] Kao, Y., He, R., & Huang, K. (2017). Deep Aesthetic Quality Assessment With Semantic Information. IEEE Transactions on Image Processing, 26(3), 1482-1495.

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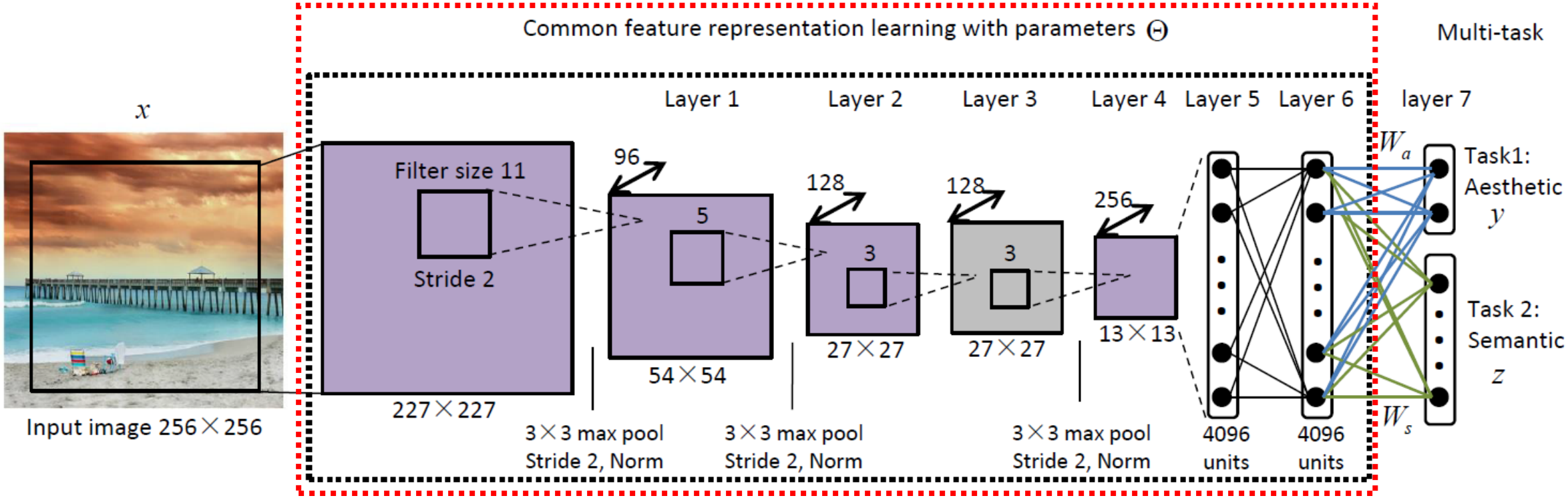

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Selected Deep Aesthetic Methods

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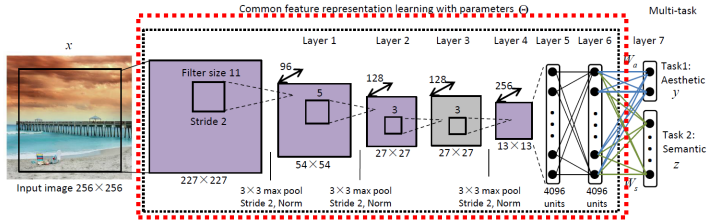
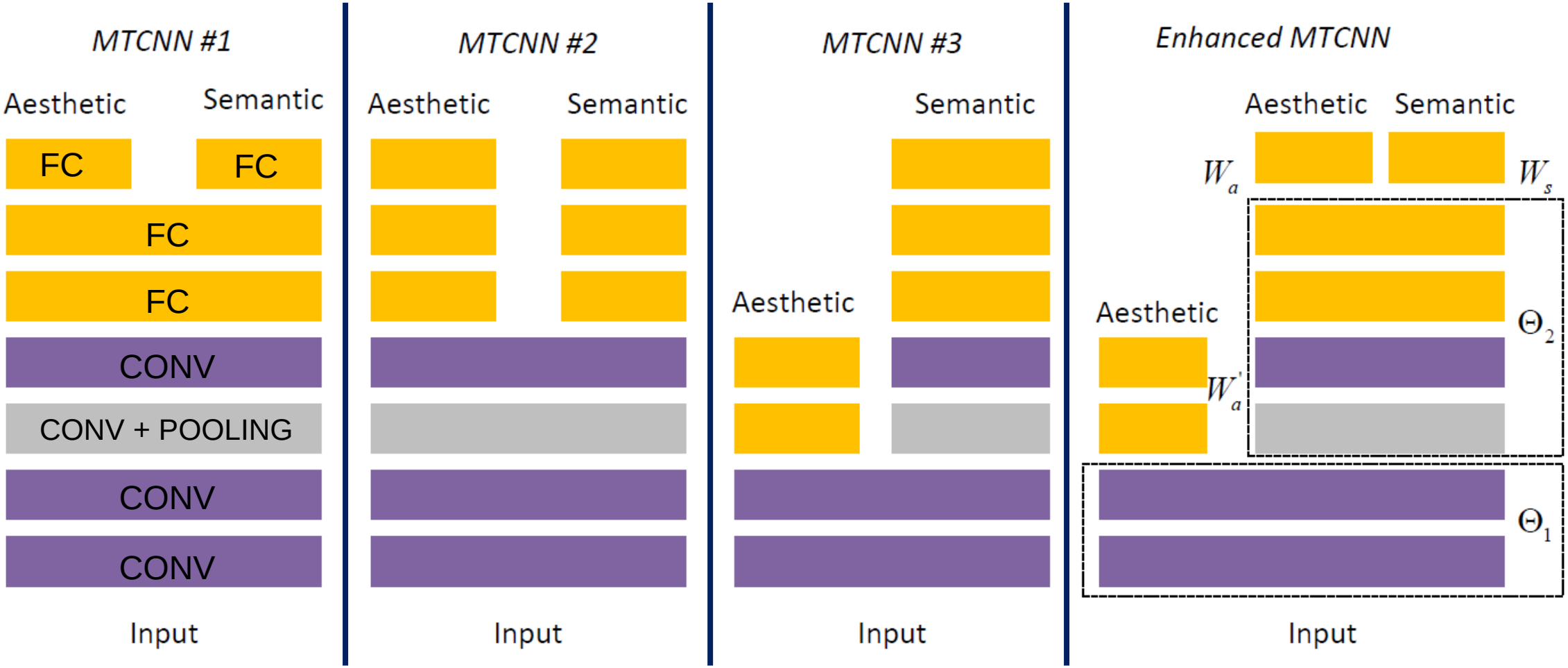
  - **Network Architecture**





Selected Deep Aesthetic Methods

- Personalized Image Aesthetics
  - Impractical to request every user to label lots of data and train a user-specific model.

[4] Ren, J., Shen, X., Lin, Z., Mech, R., & Foran, D. J. (2017, October). Personalized Image Aesthetics. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (pp. 638-647).

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    - Step 1. Train a **generic** aesthetic model. (common preference)
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    - Step 2. **Adapt** the generic model to individual users using a **limited** number of individual user's labeled examples.
    - <u>Where to get individual user's labeled examples?</u>
      - Collect **14** personal albums, each album has **~205** photos.
      - Request the **owner of the album** to rate for their own photos.

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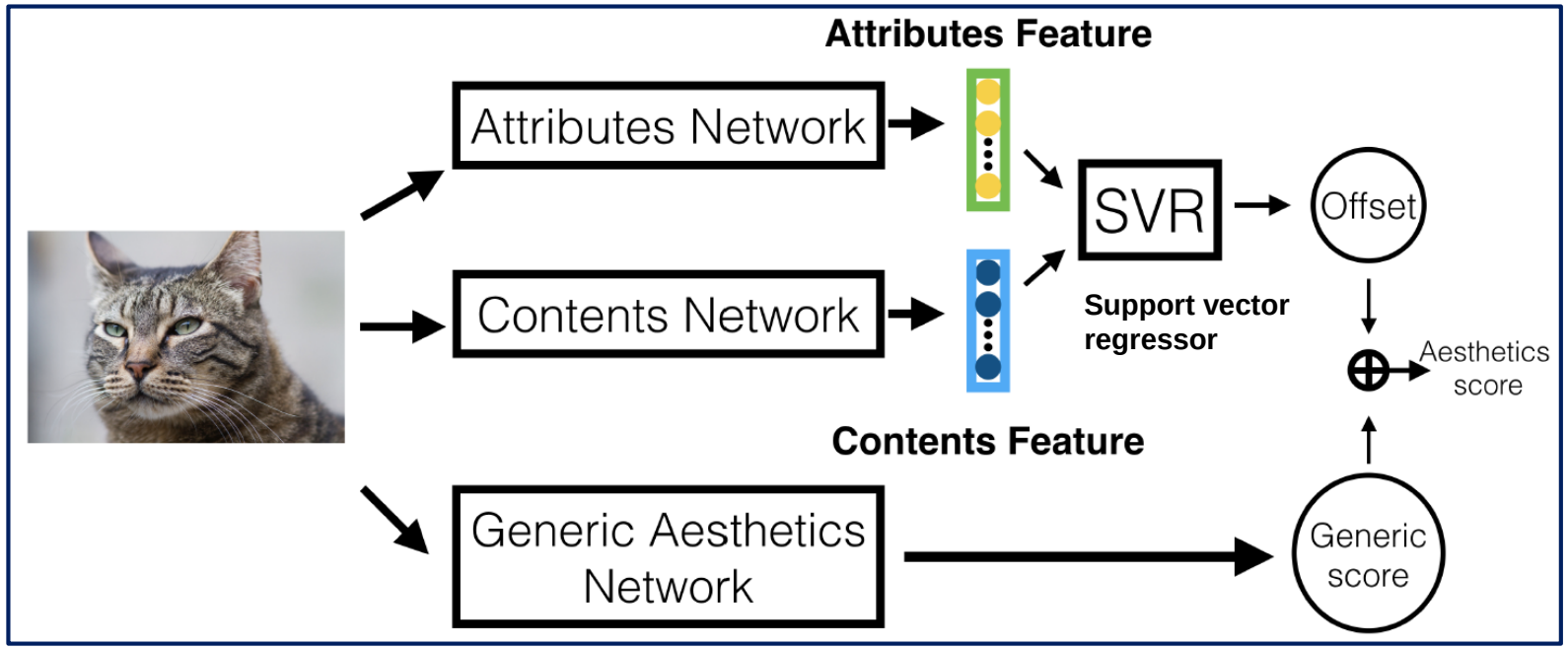
Selected Deep Aesthetic Methods

- Personalized Aesthetics Model (PAM)
  - 1) Generic aesthetic prediction
  - 2) Residual learning for personalized aesthetics
    - Learn the offset.
    - As the data is **NOT enough**, use high-level features and simply exploit Support Vector Rregressor (SVR) to do regression, instead of FC layer.

PAM Design



Selected Deep Aesthetic Methods

- Composition-preserving Deep CNN
  - Motivation: Current aesthetics algorithms typically **transform images** as pre-processing, which hurt the performance. (*Due to the FC layer to do the regression*)

(a) Cropping

(b) Scaling

(c) Padding plus scaling

Pre-processing

Selected Deep Aesthetic Methods

- Composition-preserving Deep CNN
  - Approach: Propose an adaptive spatial pooling operation.

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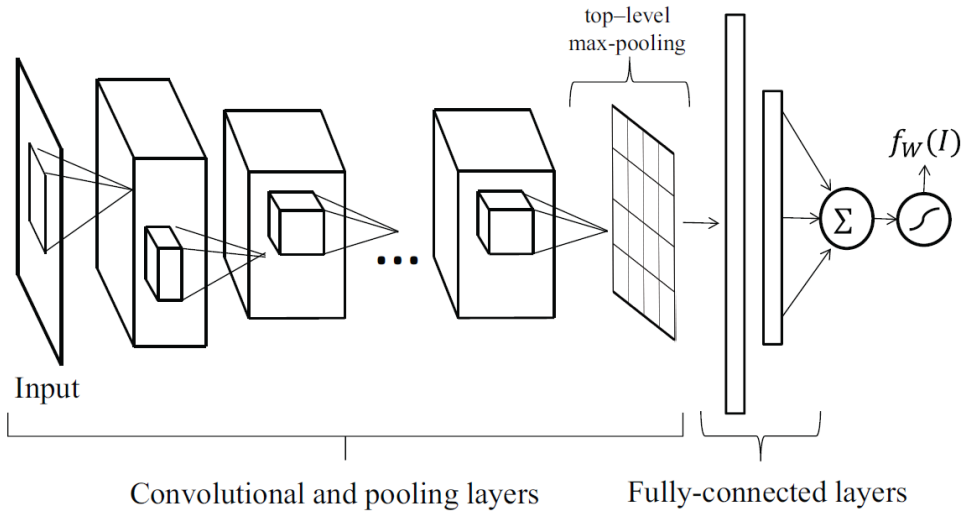

Regular Pooling

(Output feature map size **varies** with the input)

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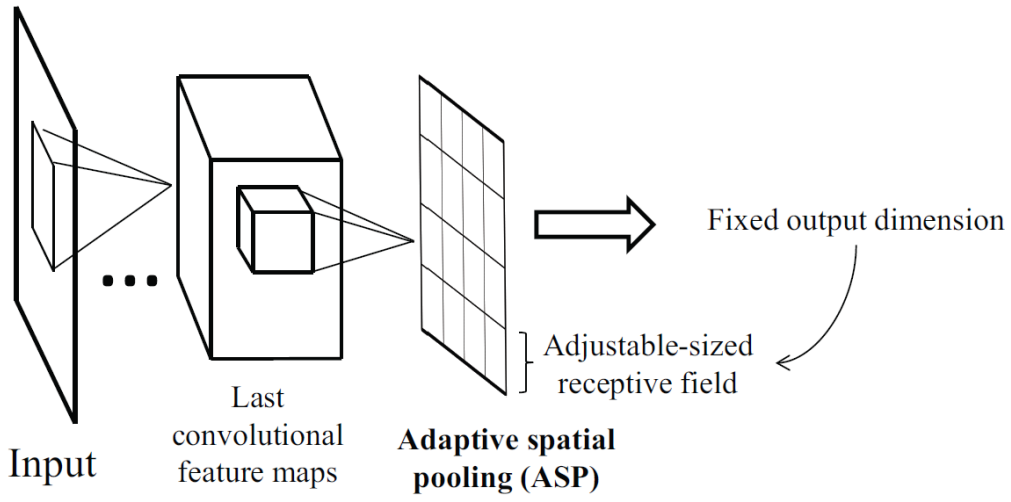
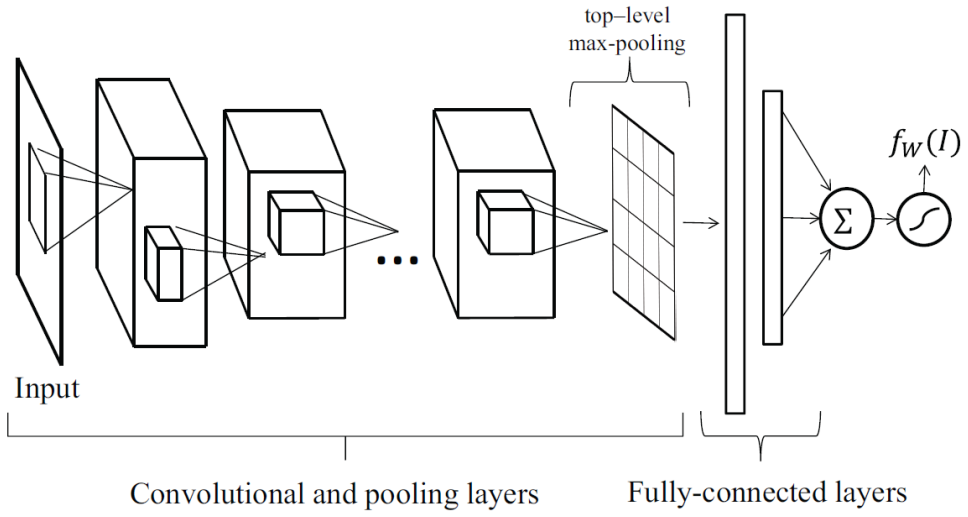


Adaptive Spatial Pooling

(Output feature map size **is fixed**)



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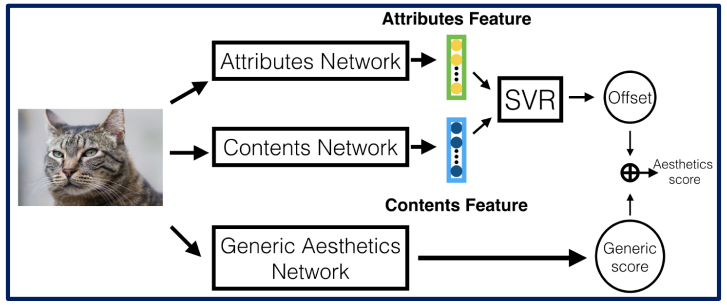
Conclusions and Future Work

- Conclusions
  - A **multi-task network** can incorporate different information and help learn the aesthetic scores.
  - **Semantic** and **attribute** information are effective in learning aesthetics scores as well as personalized aesthetics.



  - **Resizing & cropping** input images as preprocessing can hurt the performance of aesthetics prediction network.

(a) Cropping

(b) Scaling

Conclusions and Future Work

- Future Work
  - Explore self-supervised or unsupervised **task-specific** image aesthetics assessment algorithm. (not only photography aesthetics)
  - **Creating** images with high aesthetic score.
    - Already one paper from Google:

      "*Creatism: A deep-learning photographer capable of creating professional work*"

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INPUT:

OUTPUT:

Discussion

Thanks!